

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	549055
Award Program	Pollinator Health Fund
Project Title	Effects of Multiple Stressors on Pollinator Health in the Southern Plains
Grant Start Date	01/01/2018
Grant End Date	12/31/2022 (includes extension)
Total Grant Budget	\$467,416
Total FFAR Budget	\$233,708
Matching Funder(s)	Oklahoma State University Syngenta Crop Protection, LLC
Final Report Date	03/01/2024
Project Director/Principal Investigator Information	
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Stillwater, OK, 3rd

- 1.2. How many new jobs were created because of this grant funding?

Two graduate research assistant positions and one (short-term) postdoc position

- 1.3. How many jobs were maintained because of this grant funding?

None

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Bees have an important role as pollinators for flowering plants, including agricultural crops. Yet, honeybees and native bees face a variety of threats that have decreased their populations. Our research examined how different land uses, including canola, wheat, and grasslands, affected exposure of bees to pesticides, parasites, and nutrition. Specific goals included evaluating how these land use types influenced 1) the composition of the native bee community, 2) the selection of floral resources by bees, 3) the nutritional quality of bee food (i.e., pollen), 4) parasitism and disease of honeybee hives, and 5) pesticide exposure.

Thirty-six genera of bees were collected across the three land use types, with twenty-four genera occurring in all three land uses, while five genera were unique to canola, three to wheat, and two to grasslands. Canola landscapes supported a greater abundance and diversity of native bees early in the season compared to grassland and wheat landscapes, but that pattern was not observed later in the season. There was overlap between honeybees and native bees in pollen selection, especially for the most dominant pollen types early in the season. Overlap remained throughout the season, but differences were also observed in dominant, secondary, and minor pollen types. Land use type did not seem to have a large effect on the nutritional quality of pollen collected by honeybees or native bees. Although, there were some smaller differences in macronutrient collection over time among land use types. However, it is less clear how pollen availability (quantity) may have influenced bee reproductive success and health. The presence of *Varroa* mites and load scores for *Nosema* also did not vary across land use types. Pesticides were detected frequently in pollen and on plant surfaces across all land use types at similar frequencies. However, concentrations were well below acute toxic levels in nearly all samples.

Our results suggest that bees can navigate these diverse landscapes to buffer their nutritional intake and potentially limit pesticide exposure. Land use types in close proximity to native bee nesting sites and honeybee hives may be less important as long as the larger landscape has a diversity of land uses in which bees can collect pollen. Pesticide exposure may be lower in this system compared to other agroecosystems, especially since canola production has decreased in recent years as pesticide use is higher in canola than in wheat. However, that may change if canola production increases in the future.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

Canola may play an important role in supporting the native bee community early in the year. Our results suggest that canola is supporting a larger and more diverse bee community compared to wheat and grassland landscapes during the period of canola bloom, but that floral resource availability is lower later in the year. However, lack of nutritional diversity, pesticide exposure, and lack of resources later in the year may result in negative

consequences. For nutrition, our results suggest that bees were able to collect pollen with relatively similar nutritional content despite being in landscapes near quite different land uses. Similarly, low levels of pesticides in pollen collected by honeybees suggest broad and diverse foraging resulting in less exposure in hives near canola fields and higher, but still not acutely toxic, concentration in grassland sites than would be expected. Grassland, canola, and wheat landscapes were defined using the proximal land use. However, our study sites were embedded in diverse landscapes that contained fencerows and buffer zones along roads. Bees appear able to navigate these diverse landscapes to buffer their nutritional intake and potentially limit pesticide exposure. Hence, one recommendation in terms of bee nutrition is to not worry as much about the immediate land use types near native bee nesting sites and honeybee hives as long as the larger landscape has a diversity of land uses in which bees can collect pollen. Pesticide exposure may be lower in this system compared to other agroecosystems as canola production has decreased in recent years (at least partially reflecting the decision not to locate an oilseed crushing plant in Enid, Oklahoma, although that could change in the near future with a new plant opening in Goodland, Kansas in 2024) and pesticide inputs have been traditionally low in wheat.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

We identified differences in the abundance and diversity of the bee community in canola versus grassland and wheat landscapes, primarily greater bee abundance and diversity near canola fields during the period of canola bloom. We also identified overlap in pollen resource use between honeybees and native bees, especially during the period of canola bloom, which was also when resources were most abundant and likely not limiting. Our results suggest that there are not large differences in the nutritional quality of resources collected by honeybees and native bees in different land use types. However, we were not able to evaluate the overall quantity of pollen collected by bees, which could differ among landscapes. Hence, future studies may not need to focus as much on the nutritional analysis of pollen but rather focus more on the overall quantity of pollen collected.

These specific landscapes resulted in relatively low pesticide exposure to bees. The observed results may reflect recent decreases in canola production due at least partially to a lack of a nearby oilseed crushing plant. Wheat is traditionally considered a low pesticide input system, whereas canola is a higher input system. However, the opening of an oilseed crushing plant in Kansas could change these patterns if canola production increases in future years.

Our sites also did not have differences in parasite loads across landscapes. Similar parasite loads suggest similar levels of hive health, potentially reflecting the lack of differences across land use types for nutrition and pesticide exposure. A diverse landscape is likely helpful for reducing environmental impacts related to low quality or contaminated resources.

Study sites were defined by proximal land use and each site had varied land uses at spatial scales relevant to the foraging abilities of honeybees, as well as many native bee species. The lack of differences by land use categories suggests bees were utilizing resources across the larger landscape reducing potential impacts from proximal environments that are lower in floral resources and more likely to have pesticide exposure.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The overall goal of this project was to evaluate how multiple stressors influence the abundance, diversity, selection, and quality of floral resources for honeybees and native bees. Specific goals included evaluating how land use and management practices influence 1) interspecific interactions, including A) composition/structure of the native bee community (abundance and diversity), and B) parasitism/disease of honey bees, and 2) bee food quality, including A) pollen and nectar nutritional content, B) pesticide exposure via floral resources, and C) selection of floral resources by bees. These goals were addressed based on two years of field-collected data, including extensive processing of field samples in the laboratory.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No goals changed, but the timeline for completing the project was delayed due to the COVID-19 pandemic (see section 2.10).

- 2.6. What was accomplished under the goals/specific aims or milestones for the



project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

1) Major Activities

We identified 48 sampling sites for this study located in northwestern Oklahoma (including Grant, Alfalfa, Major, Garfield, and Kingfisher counties) across three different land use types: canola crop (*Brassica napus*), winter wheat crop (*Triticum aestivum*), and grassland. Site classification was based on the predominant land use of the site during the first sampling period of the study each year. Each land use type had eight replicates each year (total of 48 sites, 24 sites in 2019 and 24 different sites in 2020). Collaborators identified canola and wheat sites for the project in 2019, and grassland sites for that year were subsequently selected based on the following criteria: accessibility, proximity to canola and wheat sites, a minimum of 0.5 km from the transects (described below) of other sites, and landowner consent. No sites from 2019 were duplicated in 2020.

To determine native bee abundance and diversity at each site, we established a 150-m transect in the roadside adjacent to each field, and placed groups of pan traps (brightly colored plastic bowls used to attract bees), every 50m along the transect. Pan traps were collected after seven days, during each of three sampling periods per year. Bees present in the bowls were identified to genus level. We placed three honeybee hives at each site for approximately six months (April to September). We monitored hive weight and parasite burden (*Varroa* and *Nosema*). Macronutrient information was collected from pollen collected from honeybees entering their hives and from pollen loads carried by native bees caught in pan traps. Pollen from both sources was also identified to family and relative abundance calculated. Honeybee pollen was analyzed for a thorough list of pesticides by the USDA. Finally, to better assess pesticide applications, surface swipes were collected from leaf material at each site during each sampling period and analyzed for commonly used pesticides in the region.

2) Specific Objectives

The overall goal of this project was to evaluate how multiple stressors influence the abundance, diversity, selection, and quality of floral resources for honeybees and native bees. Specific objectives included evaluating how land use and management practices influence 1) interspecific interactions, including A) composition/structure of the native bee community (abundance and diversity), and B) parasitism/disease of honeybees, and 2) bee food quality, including A) pollen nutritional content, B) pesticide exposure via floral resources, and C) selection of floral resources by bees.

3) Major Findings

Native Bees – Abundance and Diversity

We identified 36 bee genera from pan traps. A large majority (>75%) were collected during the first sampling period when canola was in bloom. The most collected bee genus was *Lasioglossum* (63% of samples), followed by *Agapostemon* (14%), *Melissodes* (5%), *Halictus* (5%), *Augochlorella* (4%), and *Apis mellifera* (2%), with all remaining genera making up less than 1% of samples and 12 genera consisting of fewer than 5 individuals per year. Approximately 68% of bees were collected adjacent to canola fields with 60% of bees collected adjacent to canola fields collected during the first sampling period. A similar number of bees were collected adjacent to grassland and wheat sites for all sampling periods combined. The fewest bees were collected during the second sampling period, and the most bees were collected adjacent to wheat fields during the third sampling period.

Honeybees – Hive Weights and Parasite Loads

Hive weight can be a metric of overall health for honeybee hives. During the study, hives, which were not supplemented during the growing season, did not gain much mass. There was a significant difference across years (Figure 1) but not across land uses.

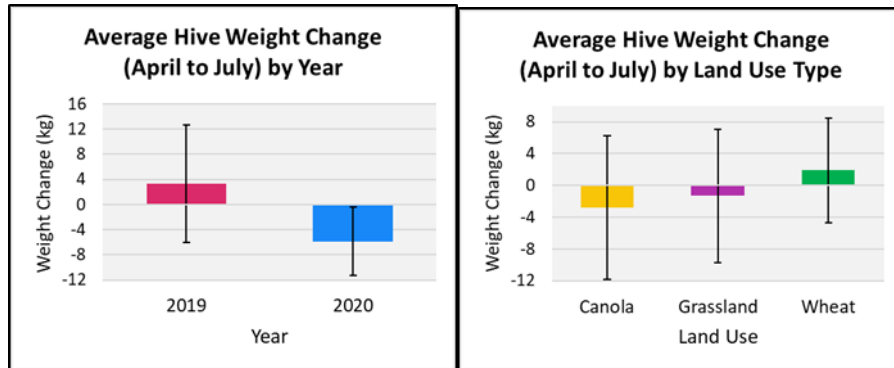


Figure 1. Hive weight changes for honeybees by year and by land use. There was a significant difference across years ($p < 0.001$), but not by land use.

We hypothesized that parasite load would be correlated to hive health as an increase in parasites could occur if hives are not healthy due to low food quality or pesticide exposure and the action of parasites could reduce the health of a hive. The presence of *Varroa* mites followed the same pattern as hive weight, with year being significant but not land use. In 2020, when hives had a net loss of weight, the number of mites was higher (Figure 2). There was not a significant difference or strong trends across year or land use for *Nosema* spores in bees.

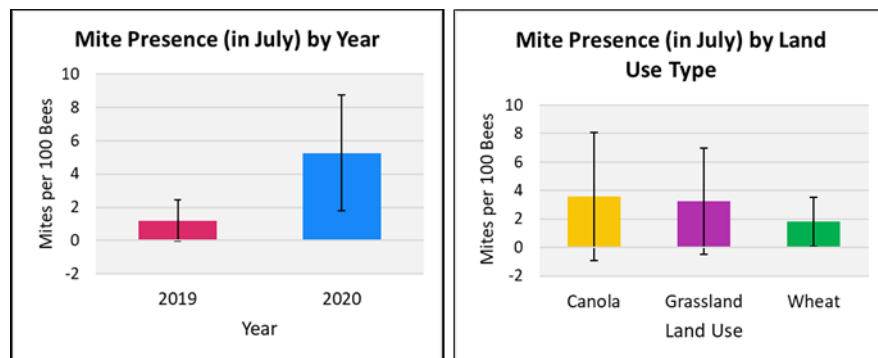


Figure 2. Parasite load (*Varroa* mites) for honeybee hives by year and by land use. There was a significant difference across years ($p < 0.001$), but not by land use.

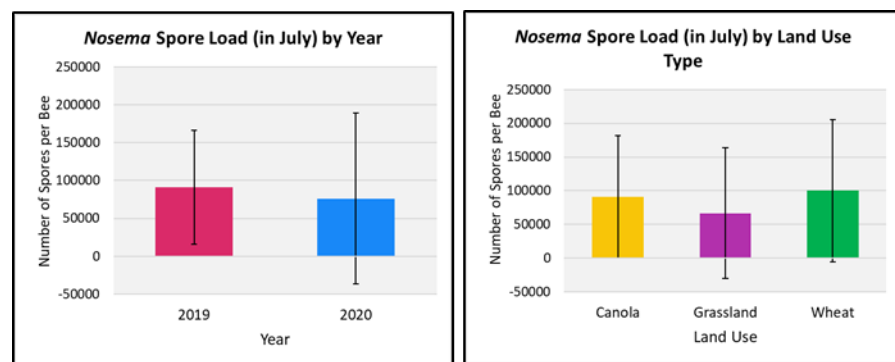


Figure 3. Parasite load (*Nosema* spores) for honeybee hives by year and by land use. There was not a significant difference across the years or by land use.

Food Quality – Nutrition

Protein content of pollen ranged from 11-30% by dry weight while carbohydrate varied from 17-34% on average, and lipid from 5-9%. Land use type did not seem to have a large effect on the nutritional quality of pollen collected by honeybees or native bees. However, there did appear to be some temporal patterns in nutrients. Protein concentrations were typically lowest during the second sampling period in 2019 while there was not a major seasonal pattern in 2020. Carbohydrate concentrations showed a slight pattern of increase from spring to fall in canola and grassland but not wheat in 2019. In 2020, carbohydrate concentrations in pollen increased from spring to fall in grasslands but peaked in summer in both canola and wheat. Lipid concentrations declined across seasons in sites in 2019 but showed a slight increase across seasons in 2020. Concentrations of most amino acids appeared to decrease from spring to fall in all three landscapes.

Food Quality – Pesticide Exposure

Pesticide exposure is greatly dependent on land-use. Based on information provided by growers and extension agents, canola is the most intensively treated landscape, especially regarding insecticides, which are most likely to be acutely toxic at environmentally relevant concentrations. Wheat also has pesticide usage including fungicides and herbicides but is considered a lower pesticide input system than canola. Grassland has the least pesticide exposure, although herbicides are sometime used. We hypothesized that landscapes that were predominately canola would have higher pesticide burdens. Pollen samples were analyzed only for honeybee hives as native bee pollen loads were too small to allow for sensitive analyses.

Pesticides in Honeybee Pollen

A total of 33 samples of honeybee pollen were analyzed for 192 pesticides. Samples included 12 from canola land use, 13 from grassland land use, and 8 from wheat land use. Of the 33 samples, 19 were from sampling period 1 (collected in April/early May), 10 from sampling period 2 (collected in July), and 4 from sampling period 3 (collected in September/early October). At least one pesticide above trace levels was detected in 21 of the 33 samples. No pesticides (at either trace or ppb levels) were detected in three of the 33 samples.

Seven pesticides were detected at trace levels only. Six pesticides were detected above trace levels but in only one sample. Seven pesticides were detected above trace levels in multiple samples (Table 1). All detection concentrations were well below 48-hour oral LD50 levels, with one exception; imidacloprid was detected at 49 ppb (a grassland site in May 2019), which is above the recorded oral LD50 level of 38 ppb.

Most pesticide detection occurred during sampling period 1 (spring), with 18 of 19 samples containing at least one pesticide above trace levels. No pesticides above trace levels were detected in samples from sampling period 3, and at least one pesticide above trace levels was detected in 3 of the 4 samples from sampling period 2.

Table 1: Pesticides that were detected above trace levels in more than one of the 33 samples. NA indicates that an Oral LD50 is not available in the literature or EPA Ecotox database.

Pesticide	Classification	Oral LD50 (ng/g)	Land Use Type	Number of Samples Detected In	Detection Range (in ppb)
Atrazine	Herbicide	970,000	Canola	6	4 - 280
			Grassland	4	50 - 101
			Wheat	4	23 - 87
Azoxystrobin	Fungicide	250,000	Canola	4	11 - 21
			Grassland	6	4 - 72
			Wheat	5	9 - 79

Carbendazim	Fungicide	NA	Canola	1	26
			Grassland	5	26 - 86
			Wheat	3	44 - 81
Chlorpyrifos	Insecticide	2500	Canola	4	5 - 101
			Grassland	4	13 - 22
			Wheat	4	13 - 65
Diuron	Herbicide	NA	Canola	1	14
			Grassland	3	5 - 14
			Wheat	2	8 - 38
Propiconazole	Fungicide	1,000,000	Canola	2	13 - 60
			Grassland	4	10 - 30
			Wheat	3	14 - 53
Tebuconazole	Fungicide	850,000	Canola	3	26 - 71
			Grassland	3	34 - 67
			Wheat	1	35

Pesticide Residue on Crop and Wildflower Surfaces

A total of 222 samples were analyzed for the 21 pesticides: 16 samples from canola plants, 19 from wheat plants (including three instances of duplicate samples from the same site and sampling period), 134 samples collected from wildflowers along transects, 29 field blanks, and 22 from crops other than canola and wheat (sorghum and soybean) that were collected during sampling periods two and three. Ten of the 21 pesticides were detected at levels above quantitation limits in the samples.

Of the 10 pesticides, seven were in eight or fewer of the 222 samples (<4% of the samples). For the insecticide flupyradifurone, the pesticide was detected in 20 (9%) of the 222 samples, with 18 of the 20 samples occurring during sampling period 1 of 2020 (35% of the sampling period 1 2020 samples). Methyl parathion occurred pervasively in the samples (60.8% of the samples, respectively). Methyl parathion occurred in 3 of the 38 samples (7.9%) from sampling period 1 in 2019, in 39 of the 65 samples (60%) from sampling periods 2 and 3 from 2019, in 38 of the 51 samples (74.5%) from sampling period 1 in 2020, and in 57 of the 68 samples (83.8%) from sampling periods 2 and 3 in 2020.

For all pesticides all concentrations were below the dermal LD50 (Table 2). All detection levels for all pesticides were below the pesticide application rates, with the exception of one sample of azoxystrobin suggesting that applications were not recent to the sampling date.

Table 2: Pesticides detected above 200 ppb in the pesticide residue surface swipe samples. NA indicates

that no dermal LD50 was available in the literature or the EPA Ecotox database.

Pesticide	Classification	Dermal LD50 (ng/cm ²)	Application Rate (ng/cm ²) Canola	Application Rate (ng/cm ²) Wheat	Number of Samples Detected In	Average $\pm$ Standard Deviation (in ng/cm ²)	Detection Range (ng/cm ²)
Ethalfuralin	Herbicide	NA	10,500	10,500	3	6.86 $\pm$ 0.82	6.09 - 7.72
Atrazine	Herbicide	92,000	19,800	19,800	7	21.33 $\pm$ 33.87	5.38 - 97.93
Methyl Parathion	Pesticide	157	128000	13340	137	15.72 $\pm$ 9.05	5.02 - 43.24
Chlorantraniliprole	Insecticide	2.19E+16	2240.00	2240	1		9.79
Propiconazole	Fungicide	238,000	1,250	1,250	5	29.37 $\pm$ 48.74	2.80 - 115.62
lambda Cyhalothrin	Insecticide	210	336.00	336	8	7.35 $\pm$ 8.28	2.84 - 26.86
Flupyradifurone	Insecticide	131000	282.00	4090	20	8.70 $\pm$ 2.53	5.92 - 15.34
Zeta Cypermethrin	Insecticide	19	26.00	129	3	8.71 $\pm$ 3.27	6.73 - 12.48
Azoxystrobin	Fungicide	190,000.00	642.00	147.00	2		107.77 - 172.22

Food Quality – Floral Resource Selection by Bees

During the first sampling period, Brassicaceae was a major pollen type for all honeybee hives located in canola landscapes, and most of the hives located in grassland and wheat landscapes. Later in the season, Chenopodiaceae and Euphorbiaceae were major pollen types across land use types, with other pollen types being important for some hives in grassland or wheat landscapes, including Fabaceae. During the third sampling period, Asteraceae was important across land use types.

Brassicaceae was also important to native bees early in the season, as well as Fabaceae and Asteraceae, and Amaryllidaceae was important for some bees in grassland or wheat landscapes. Asteraceae and Fabaceae continued to be important across land use types during the second sampling period, as well as Convolvulaceae and Euphorbiaceae for some individuals. Asteraceae was the most important pollen type across land use types during the late sampling period. Fewer bees were collected and thus pollen loads evaluated during the third sampling period.

4) Key Outcomes

All key outcomes are described in previous sections, including implications of the results for producers and future research (see sections 2.2 and 2.3).

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

In addition to Standard Operating Procedures for field and laboratory procedures, we also had some samples analyzed through other labs. We had a subset of the pollen samples analyzed for protein content using amino acid analysis in a laboratory run by a Missouri State chemist. Amino acid analysis is considered to be one of the most accurate measures of protein content and has the additional benefit of providing data on the concentrations of individual amino acids in samples. For pesticide analyses, standard quality control procedures were followed including trip blanks, laboratory blanks, laboratory control samples, internal standards, and continuing instrument calibration. For honeybee hive pollen, samples were sent to the U.S. Department of Agriculture's National Science Laboratories (Gastonia, NC) for pesticide analysis. We also utilized reference collections for bee and pollen identifications.

- 2.8. List key stakeholders that could be served by results of this project.

Stakeholders who could be served by the results of this project include beekeepers, canola producers, wheat producers, cattle producers, and grassland managers, although there is extensive overlap among these categories (i.e., producers are often diversified) in this region.

- 2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

We have communicated with collaborating producers throughout the duration of the project,

including providing basic information about their sites. We are looking for additional opportunities to disseminate outcomes more broadly, such as through incorporation into extension fact sheets.

- 2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

We previously experienced delays due to COVID-19 which were described in annual reports. Graduate student progress continued to be delayed throughout the duration of the project, partly due to the continued effects of the pandemic and partly due to other extenuating circumstances.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

We provided training for two Ph.D. students throughout the duration of the project, and for a short-term postdoc. Students were included in all aspects of the scientific process. One student focused on the nutritional aspects of the project and one student focused on the pesticide aspects of the project. The students received training in pollinator fieldwork (e.g., pan traps, pollen traps, pesticide residue collection), pollen identification, bee identification, and the analysis of pesticide residues and macronutrient content of pollen. The postdoc focused on native bee identifications. Graduate students submitted annual reviews that summarized their activities each year, including goals, plans, and professional development activities. Graduate students and the postdoc took advantage of various professional development opportunities over the duration of the project, including micro-credentials available through OSU's Graduate College, other opportunities available through Aurora and the National Center for Faculty Development and Diversity, and activities through regular lab group meetings, as well as one-on-one mentoring with the PIs.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Two Ph.D. students and a short-term postdoc participated in this project.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

Scholarly publications and dissertations resulting from this project are in progress.

- 3.2. Please provide a list of citations for the information products produced during the project.

See request in the next section (3.3).

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

If there is a process to do so, we will provide publications as they become available.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

Research activities are not currently publicly available via websites.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No inventions or patent applications have resulted from this project.

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
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Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Specific site locations are not provided to protect landowner privacy. All study sites were located on private land.

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?

The data collected for this project are the basis for two dissertation projects. The two Ph.D. students are nearing completion, but they have both experienced delays for various reasons. Following completion and subsequent publication (within a reasonable timeframe), data will be publicly available. We anticipate all data will be included in peer-reviewed publications. However, if any data sets should remain unpublished, they will be described on a separate page of the PI's webpages and available for sharing to those who request access.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Data generated focused on native bees (1- community composition to genus; 2- floral composition of pollen loads; 3- nutritional composition of pollen loads (% protein, % carbohydrates, % lipids)) and honey bees (1- hive weights; 2- *Nosema* and *Varroa* levels; 3- floral composition of pollen loads (with major and secondary pollen types by family); 3-

nutritional composition of pollen loads (% protein, % carbohydrates, % lipids; amino acids);
4- pesticide analysis of pollen loads), as well as environmental samples (pesticide analysis of
leaf surface swipes) and LD50 references (both contact and oral).

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide
ashort description of how they are related.

All datasets are related by year, sampling period, site, land use type. Native bee samples also
include pan trap information. Honeybee samples also include hive number.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission
ofenhancing the discoverability of FFAR funded project data and information
products. Include or attach copies of records and a simple file inventory, if
necessary, in a compressed folder.

It wasn't possible to submit an excel file, so metadata is provided as an attachment here
(see following pages), including a table of contents and a separate page(s) with metadata
for each dataset.

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Grant ID	549055
Award Program	Pollinator Health Fund
Project Title	Effects of Multiple Stressors on Pollinator Health in the Southern Plains
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Project Director	Kristen Baum
Awardee Institution	Oklahoma State University
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Datasets	Datasets are listed in alphabetical order and metadata for each dataset are provided on a separate page.
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Honeybee Hive Weights

Dates of Sample Collection	March-September 2019; April-October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Sampling Protocol	Three honey bee hives were placed at each site. Approximately once per month, hives were weighed in kilograms using a digital electronic scale.
A - Year	Year of sample's collection
B - Site	The site ID for where the sample was collected from
C - LandUse	The proximal land use type for the site that the pollen was collected at; what land use (canola crop, wheat crop, or grassland) was surrounding or nearest to the hives
D - HiveID	Unique ID assigned to each hive in the study
E - HiveSource	The source of the hive (either a nuc from Bee Weaver; a daughter hive that's been split from a hive purchased the previous year; a hive from the previous year that had a second hive split from it [mother hive]; or a hive from the previous year that was not split). Numbers are used to reference the hive ID number used the previous year.
	For all columns following E, *means no weight was collected for weight columns; n/a means no date is provided in the date column because no weight was collected; dead in a weight column means that no living bees were found in the hive; stolen in a weight column means the hive was removed from the site by an outside party; see notes indicates a unique situation prevented data collection
F - March Weight	The weight in kilograms of the hive in March
G - MW Date	The date that the March weight was measured
H - April Weight	The weight in kilograms of the hive in April
I - AW Date	The date that the April weight was measured
J - May Weight	The weight in kilograms of the hive in May
K - MW Date	The date that the May weight was measured
L - Late May Weight	The weight in kilograms of the hive in late May
M - Late May Date	The date that the late May weight was measured
N - June Weight	The weight in kilograms of the hive in June
O - JuneW Date	The date that the June weight was measured
P - July Weight	The weight in kilograms of the hive in July
Q - JulyW Date	The date that the July weight was measured
R - Sept Weight	The weight in kilograms of the hive in September
S - SeptW Date	The date that the September weight was measured
T - Notes	Any relevant details about the status of the hives during any of the days the the weight was measured or additional information to explain why the hive weight was not measured

Honeybee Nosema

Dates of Sample Collection	July 29th-30th 2019, July 31st 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Sampling Protocol	Three honey bee hives were placed at each site. During each sampling period, 1/2 cup of bees (approximately 300 bees) were collected from each hive and preserved in 80% ethanol. For each sample during sampling period 2 (July), 60 bee abdomens were crushed in 60 mL of water, and 10 μ L of the slurry was immediately added to a hemocytometer slide. The number of spores in 80 of the smallest squares were then counted.
A - Year	Year of sample's collection
B - SamplingPeriod	The three times of the year in which pollen was collected; 1 was during late April/early May; 2 was during July, and 3 was during September/early October
C - Site	The site ID for where the sample was collected from
D - LandUse	The proximal land use type for the site that the pollen was collected at; what land use (canola crop, wheat crop, or grassland) was surrounding or nearest to the hives
E - HiveID	Unique ID assigned to each hive in the study
F - Bees	The number of honey bees in the sample
G - Sporesonslidegrid	The number of spores counted in the 80 squares on the hemocytometer slide
H - Sporesperbee	The number of spores per bee in the sample, based on the calculation provided by Cantwell (1970): (Total number of spores counted/Number of squares counted)*(4×10^6)

Honeybee Pollen Amino Acids

Dates of Sample collection	May - September 2019; April - October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Honey Bee Sampling Protocol	Three honey bee hives were placed at each site. During each sampling period, plastic pollen traps were placed on the entrance of each hive. Pollen samples were transferred to glass vials and frozen until processing. Pollen samples were homogenized and part of the sample was used for nutrient analysis and part of the sample was used for pollen identification.
Amino acid analysis	Analysis of the amino acid content of subsamples of pollen was done by the Agricultural Experiment Station Chemical Laboratories at the University of Missouri-Columbia
A - Landscape	The proximal land use type for the site (canola crop, wheat crop, or grassland)
B - Sampling Period	Samples were collected three times per year, including late April/early May (sampling period 1), July (sampling period 2), and September/early October (sampling period 3).
C - Alanine	Each of these variables represents the percent dry mass of the sample composed of each of the named amino acids
D - Arginine	
E - Aspartic Acid	
F - Glutamic Acid	
G - Glycine	
H - Histidine	
I - Isoleucine	
J - Leucine	
K - Lysine	
L - Methionine	
M - Phenylalanine	
N - Proline	
O - Serine	
P - Threonine	
Q - Tyrosine	
R - Valine	
S - Total amino acid	The total percent of the sample composed of the measured amino acids

Honeybee Pollen Nutrients

Dates of Sample collection	May - September 2019; April - October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Honey Bee Sampling Protocol	Three honey bee hives were placed at each site. During each sampling period, plastic pollen traps were placed on the entrance of each hive. Pollen samples were transferred to glass vials and frozen until processing. Pollen samples were homogenized and part of the sample was used for nutrient analysis and part of the sample was used for pollen identification.
Pollen Nutritional Analysis	Pollen protein content was quantified using the Bradford Assay modified for microplate use, and preceded by digestion of samples in 0.1 M NaOH. A subsample of this digested sample was also used to quantify carbohydrate concentration using the Anthrone assay. Pollen lipid content was quantified on a separate subsample using the phospho-vanillin assay.
A - Landscape	The proximal land use type for the site (canola crop, wheat crop, or grassland)
B - Year	The year of the sample's collection
C - Sampling Period	Samples were collected three times per year, including late April/early May (sampling period 1), July (sampling period 2), and September/early October (sampling period 3).
D - Protein (%)	Percent protein content of a sample on a dry matter basis
E - Carbohydrates (%)	Percent carbohydrate content of a sample on a dry matter basis
F - Lipids (%)	Percent lipid content of a sample on a dry matter basis

Honeybee Pollen Pesticides

Dates of Sample Collection	April to September 2019 and April to October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Sampling Protocol	Three hives were placed at each site. Plastic pollen traps were attached to the entrances honey bee hives for four days. The contents of the traps were collected and then manually homogenized by site for each sampling period. Approximately 3 grams (wet weight) of pollen from each site, for each sampling period, was mailed to the USDA's National Science Laboratories (Gastonia, NC) for analysis for 193 pesticides.
A - Year	Year of sample's collection
B - Sampling Period	The three times of the year in which pollen was collected; 1 was during late April/early May; 2 was during July, and 3 was during September/early October
C - Site	The site ID for where the sample was collected from
D - HivesIncluded	Which of the hives from the site were included in the sample; hives were labeled 1, 2, and 3 at each site
E - NumHivesIncluded	How many hives (from the site) were included in the sample
F - LandUse	The proximal land use type for the site that the pollen was collected at; what land use (canola crop, wheat crop, or grassland) was surrounding or nearest to the hives
G - NumberDetectedatPPBlevels	The number of pesticides the were detected at parts per billion [PPB] levels (above trace levels) in the pollen sample
H through AA	Column title is the pesticide tested (for presence and concentration). The number in the cell below indicates the amount that was detected (in PPB [parts per billion]) in the sample that is listed for the corresponding row. Trace indicates that the pesticide was detected in the sample at levels below PPB. ND indicates that the pesticide was not detected in the sample.

Honeybee Varroa

Dates of Sample Collection	2019: May 10th-11, July 29th-30th, September 26th and 29th; 2020: May 1st-2nd, July 31st, October 6th and 8th
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Sampling Protocol	Three honey bee hives were placed at each site. During each sampling period, 1/2 cup of bees (approximately 300 bees) were collected from each hive and preserved in 80% ethanol in a mason jar. To remove the mites from the bees, each jar was shaken vigorously for approximately 30 seconds, and the contents were strained through a wire screen. The strained ethanol was inspected for mites, and mites were counted and collected. (The straining and counting procedure was repeated until the strained ethanol contained no mites.)
A - Year	Year of sample's collection
B - SamplingPeriod	The three times of the year in which pollen was collected; 1 was during late April/early May; 2 was during July, and 3 was during September/early October
C - Site	The site ID for where the sample was collected from
D - LandUse	The proximal land use type for the site that the pollen was collected at; what land use (canola crop, wheat crop, or grassland) was surrounding or nearest to the hives
E - HiveID	Unique ID assigned to each hive in the study
F - Mites	The number of mites found in the sample
G - Bees	The bee sample size for the sample

LD50 References

Contact LD50 Reference Tab	
Column	Description
A - Pesticide	Pesticide that LD50 and application rate information is provided for
B - Contact LD50 (ng/cmsq)	The contact LD50 value for honey bees (<i>Apis mellifera</i>) measured in nanograms of active ingredient per centimeter squared of bee. When calculations were required, we assumed the surface area of a bee as 1.05 centimeters squared, based on research by Poquet et al. (Citation: Poquet, Y. et al. (2014). A pragmatic approach to assess the exposure of the honey bee (<i>Apis mellifera</i>) when subjected to pesticide spray. <i>PLoS ONE</i> , 9(11), e113728.) *indicates that no contact LD50 was located in the literature
C - Application Rate (ng/cmsq) Canola	Application rate of the pesticide to canola crops measured in nanograms of active ingredient per centimeters squared
D - Application Rate (ng/cmsq) Wheat	Application rate of the pesticide to wheat crops measured in nanograms of active ingredient per centimeters squared
E - Source for Application Rate	Website link to the source of the application rate
F - Source for Contact LD50	The source for the contact LD50. List of Sources: EPA Ecotox Database (https://cfpub.epa.gov/ecotox/search.cfm); Danka, R. et al. (1986). Comparative toxicities of four topically applied insecticides to Africanized and European Honey Bees (Hymenoptera: Apidae). <i>Journal of Economic Entomology</i> , 79, 18-21.; Abrol, D. and Andotra, R. (2003). Relative toxicity of some insecticides to <i>Apis mellifera</i> L. <i>Journal of Asia-Pacific Entomology</i> , 6(2), 235-237.; Stoner, K. and Eitzer, B. (2013). Using a hazard quotient to evaluate pesticide residues detected in pollen trapped from honey bees (<i>Apis mellifera</i>) in Connecticut. <i>PLoS ONE</i> , 8(10), e77550.; Zhu, Y. et al. (2015). Spray toxicity and risk postential of 42 commonly used formulations of row crop pesticides to adult honey bees (Hymenoptera: Apidae). <i>Journal of Economic Entomology</i> , 108(6), 2640-2647.
Oral LD50 Reference Tab	
Column	Description
A - Pesticide	Pesticide that LD50 information is provided for
B - Oral LD50 (ng/g)	The oral LD50 value for honey bees (<i>Apis mellifera</i>) measured in nanograms of active ingredient per gram of bee. When calculations were required, we assumed the weight of the bee to be 0.1 g. *indicates that no oral LD50 was located in the literature

C - Source of Oral LD50	The source for the oral LD50. List of Sources: EPA Ecotox Database (https://cfpub.epa.gov/ecotox/search.cfm); Atkins, E. and Kellu, D. (1986). Comparative morphogenic and toxicity studies on the effect of pesticides on honeybee brood. <i>Journal of Apicultural Research</i>, 25(4), 242-255.; Helmer, S. et al. (2015). Effects of realistic doses of atrazine, metolachlor, and glyphosate on lipid peroxidation and diet-derived antioxidants in caged honey bees (<i>Apis mellifera</i>). <i>Enrionmental Science and Pollution Research</i>, 22, 8010-8021.; Ostiguy, N. et al. (2019). Honey bee exposure to pesticides: a four-year nationwide study. <i>Insects</i>, 10 , 13.
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Native Bee Community

Dates of Sample Collection	March-September 2019; April-October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Sampling Protocol	Four sets of pan traps were placed along a 150m transect located along the roadside adjacent to each site. Each pan trap set consisted of three traps (one fluorescent yellow, one fluorescent blue, and one white) and sets were placed every 50m along the transect. Each pan trap was filled with 1:1 mixture of propylene glycol and water and a few drops of
A - Year	Year of sample's collection
B - SamplingPeriod	Samples were collected three times per year, including late April/early May (sampling period 1), July (sampling period 2), and September/early October (sampling period 3).
C - Site	The site ID for where the sample was collected from
D - LandUse	The proximal land use type for the site (canola crop, wheat crop, or grassland)
E - PanTrapLocation	Location of pan trap based on distance along transect
F - PanTrapColor	Color of pan trap (blue, yellow, or white)
G - BeeNumber	Unique ID assigned to each bee in a pan trap
H - Family	Family of identified bee
I - Subfamily	Subfamily of identified bee
J - Tribe	Tribe of identified bee
K - Genus	Genus of identified bee; deteriorated = bee was in too poor of a condition to identify; no bees = no bees were collected in the pan trap; unknown = bee could not be identified to genus
L - Notes	Any relevant details about the status of the hives during any of the days the the weight was measured or additional information to explain why the hive weight was not measured

Native Bee Pollen Nutrients

Dates of Sample collection	May - September 2019; April - October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Native Bee Sampling Protocol	Four sets of pan traps were placed along a 150m transect located along the roadside adjacent to each site. Each pan trap set consisted of three traps (one fluorescent yellow, one fluorescent blue, and one white) and sets were placed every 50m along the transect. Each pan trap was filled with 1:1 mixture of propylene glycol and water and a few drops of Dawn dish soap. Samples were collected after one week and placed in vials with 80% ethanol until being processed for identification. A subset of the more common bee genera were selected for pollen identification. Flowers were collected from representative plants in the area where bees were collected. Once pollen had been identified from the native bees, we extracted pollen from those same families of plants and used it for nutritional analysis.
Pollen Nutritional Analysis	Pollen protein content was quantified using the Bradford Assay modified for microplate use, and preceded by digestion of samples in 0.1 M NaOH. A subsample of this digested sample was also used to quantify carbohydrate concentration using the Anthrone assay.
A - Landscape	The proximal land use type for the site (canola crop, wheat crop, or grassland)
B - Native Bee	The genus of native bee from which pollen was identified
C - Site	One of the 24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
D - Sampling Period	Samples were collected three times per year, including late April/early May (sampling period 1), July (sampling period 2), and September/early October (sampling period 3).
E - Family	The family of plants from which pollen was collected for nutritional analysis
F - Protein (%)	Percent protein content of a sample on a dry matter basis
G - Carbohydrate (%)	Percent carbohydrate content of a sample on a dry matter basis

Pollen Composition

Dates of Sample Collection	March-September 2019; April-October 2020
Sites and locations	24 sites (12 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Honey Bee Sampling Protocol	Three honey bee hives were placed at each site. During each sampling period, plastic pollen traps were placed on the entrance of each hive. Pollen samples were transferred to glass vials and frozen until processing. Pollen samples were homogenized and part of the sample was used for pesticide analysis and part of the sample was used for pollen identification.
Native Bee Sampling Protocol	Four sets of pan traps were placed along a 150m transect located along the roadside adjacent to each site. Each pan trap set consisted of three traps (one fluorescent yellow, one fluorescent blue, and one white) and sets were placed every 50m along the transect. Each pan trap was filled with 1:1 mixture of propylene glycol and water and a few drops of Dawn dish soap. Samples were collected after one week and placed in vials with 80% ethanol until being processed for identification. A subset of the more common bee genera were selected for pollen identification.
Pollen Staining Protocol	Pollen was stained with Safranin O, homogenized, transferred to a glass slide, and the first 200 encountered grains were identified.
A - Year	Year of sample's collection
B - SamplingPeriod	Samples were collected three times per year, including late April/early May (sampling period 1), July (sampling period 2), and September/early October (sampling period 3).
C - Site	The site ID for where the sample was collected from
D - LandUse	The proximal land use type for the site (canola crop, wheat crop, or grassland)
E - BeeGenus	Genus of bee from which pollen load was collected
F - Family1	Family of most abundant pollen type
G - Percent1	Percent of 200 grain pollen count for pollen type in column F
H - Family2	Family of second most abundant pollen type
I - Percent2	Percent of 200 grain pollen count for pollen type in column H
J - Family3	Family of third most abundant pollen type
K - Percent3	Percent of 200 grain pollen count for pollen type in column J
L - Family4	Family of fourth most abundant pollen type
M - Percent4	Percent of 200 grain pollen count for pollen type in column L
N - Family5	Family of fifth most abundant pollen type
O - Percent5	Percent of 200 grain pollen count for pollen type in column N
P - Family6	Family of sixth most abundant pollen type
Q - Percent6	Percent of 200 grain pollen count for pollen type in column P
R - Family7	Family of seventh most abundant pollen type
S - Percent 7	Percent of 200 grain pollen count for pollen type in column R

Surface Swipes

Dates of Sample Collection	April to September 2019 and April to October 2020
Sites and locations	48 sites (24 each year) in Grant, Alfalfa, Garfield, Major, and Kingfisher counties in Oklahoma, USA
Sampling Collection Protocol	At each site, during each sampling period, the pesticide surface residue samples were collected by dampening a 55 mm diameter filter paper with 80% pesticide grade ethyl acetate. The dampened paper was wiped across 40 centimeters squared of flowering plant leaves along a 150 meter transect located along the road at each site. For all sites, during all sampling periods, the pesticide residue samples were collected for wildflowers. For sites proximal to canola crops or wheat crops, during the first sampling period, a surface residue sample was also collected (by walking 10 meters into the crop field, from the transect center, wiping approximately 13 centimeters squared of flowering plant leaf material (located towards the top and middle of the plant), and then swiping approximately 13.5 centimeters squared from flowering plant leaves 15 meters to the left and 13.5 centimeters squared from flowering plant leaves 15 meters to the right of the center spot. During sampling periods 2 and 3, if a new crop was growing in the original crop site, a surface residue swipe was collected in that field, using the same pattern as collection for crops during sampling period 1. For each day of sample collection, an unswiped blank sample (dampened filter paper that was not swiped). A blank sample for calculating the quantitation limits were created in the lab, consisting of 62.5 microliters of internal standard and 12 milliliters of ethyl acetate, evaporated to 1 milliliter. This sample was analyzed periodically during GC-MS analysis runs of the other samples.
Sample Processing Protocol	Pesticides on samples were extracted by adding 62.5 microliters of an internal standard (consisting of 4 microliters per milliliter of polycyclic aromatic hydrocarbon internal standard and 4 microliters per milliliter atrazine D5 in acetonitrile) to glass vials containing the filter paper and eluting with 12 mL of pesticide grade ethyl acetate. Vials were vortexed for 30 seconds and evaporate to 1 milliliter with nitrogen gas. Extractions were analyzed for detection and concentration of 21 pesticides using gas chromatography-mass spectrometry.
SampleConcentrations Tab	
Column	Description
A - SampleNumber	Unique ID number for each sample
B - Year	Year of sample's collection
C - Sampling Period	The three times of the year in which pollen was collected; 1 was during late April/early May; 2 was during July, and 3 was during September/early October
D - LandUse	The proximal land use type for the site that the pollen was collected at; what land use (canola crop, wheat crop, or grassland) was surrounding or nearest to the hives.
E - SiteSampleTypeID	The site ID for where the sample was collected from
F - Site	The site ID for where the sample was collected from

G - SampleType	The source of the sample (wildflowers, canola crop, wheat crop, a new crop during sampling period 2 [sorghum, soybean, or unspecified SP2Crop if crop could not yet be identified] and field collected blank)
H through AB	Column title is the pesticide tested (for concentration). The number in the cell below indicates the amount that was detected (in PPB [parts per billion]) in the sample that is listed for the corresponding row. Only concentrations above 200 nanograms per milliliter (parts per billion [PPB]) were recorded. (* indicates pesticide was either not detected or the concentration was below 200 PPB. Red text indicates that the ion detection percentages did not match that of the compound.)
AC - Comments	Additional comments relating to the sample
QL Tab	
A - LabBlankSampleID	Unique ID number for the sample run of the lab blank sample
B through V	Column title is the pesticide tested (for concentration). The number in the cell below indicates the amount that was detected (in PPB [parts per billion]) in the sample that is listed for the corresponding row. Below the values in each column are the average, standard deviation, and calculated quantitative limit for the pesticide (calculated as the average plus 10x the standard deviation).